ACADEMIC DEFENSE & TECHNICAL EVALUATION

CuraCascade V2: Automated Discrepancy Auditing & Evidence Synchronization Across Clinical Genomics Databases

A 5-stage deterministic framework closing the multi-year evidence propagation lag between authoritative ClinGen gene downgrades and downstream ClinVar variant assertions & PanelApp diagnostic panels.

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Live Portal: curacascadev2.streamlit.app

Repository: github.com/uditshokeen14/curacascadeV2

Executive Summary: Problem, Solution & Verified Outcomes

A high-level synthesis of the clinical challenge, software solution, and empirical validation.

THE PROBLEM

Evidence Synchronization Disconnect

When ClinGen downgrades a gene from causal to Disputed, Refuted, or Limited, downstream ClinVar variants and PanelApp panels do not automatically update.

Stale Pathogenic assertions persist for an overall median of 2.82 years (up to 9.9 years), risking false-positive diagnoses and unwarranted surgeries.

THE SOLUTION

Deterministic 5-Stage Audit Pipeline

CuraCascade ingests multi-source data, normalizes disparate disease ontologies via a 3-tier MONDO-to-MedGen crosswalk, and stores data in a 3NF SQLite warehouse + DuckDB OLAP engine.

Materializes Class A, B, and C discrepancies with actionable clinical remediation directives.

STRONGEST RESULT

100.0% Precision Across 94 Benchmark Cases

Evaluated on a 100-gene independent expert adjudication benchmark (94 evaluable cases):

- Zero False Positives (100% PPV, 95% CI: [82.4%, 100.0%]).
- Zero false alarms on 42 definitive controls.
- Scaled to Full Corpus: 51,251 findings across 3,033 genes and 51,025 variants.

BIGGEST LIMITATION

34.6% Recall & Static Data Snapshots

The benchmark recall is 34.6% (18/52 detected). This reflects an intentional clinical priority: avoiding alert fatigue via conservative disease matching.

Auditing the 34 false negatives proves that 47% were absent from the snapshot, and 41% were conservative phenotypic exclusions.

02 // THE CLINICAL CHALLENGE

The Evidence-to-Practice Gap in Clinical Genomics

Authoritative gene downgrades fail to propagate to diagnostic databases, risking patient harm.

1. SCIENTIFIC RE-APPRAISAL

In early candidate-gene literature, small sample sizes and lack of population controls led researchers to falsely associate genes with severe diseases.

Today, expert consortia like ClinGen systematically re-evaluate validity, downgrading candidate genes to Disputed, Refuted, or Limited (e.g., SCN1B, MYLK2, KCNE2).

2. ASYNCHRONOUS DRIFT

Downstream databases operate under independent governance with no automated synchronization:

- **NCBI ClinVar:** Archival repository; does not re-classify variants when upstream genes are refuted.
- **PanelApp:** Diagnostic gene panels lag behind published expert consensus.

Result: 51,251 unpropagated discrepancies in our warehouse snapshot.

3. TANGIBLE PATIENT HARM

When diagnostic laboratories rely on stale assertions, real patients suffer severe clinical harms:

- **False-Positive Diagnoses:** Healthy individuals labeled as harboring lethal cardiac or cancer risks.
- **Invasive Surgeries:** Unnecessary implantable defibrillator (ICD) placements or organ resections.
- **Trauma in Families:** Inappropriate cascade testing of children.

Biomedical Foundations I: DNA, Genes, Variants & Proteins

Understanding how genetic variation causes human disease from first principles.

DNA & Chromosomes

- DNA is a chemical double-helix encoded by four chemical bases: A, C, G, and T.
- Human cells package DNA into 23 pairs of chromosomes (46 total).
- Think of chromosomes as large reference volumes containing recipes for life.

Genes & Proteins

- A gene is a specific sequence of DNA bases providing instructions to build a protein.
- Humans have ~20,000 genes.
- Proteins perform cellular work: building muscle (MYH7), conducting heart currents (SCN5A), or repairing DNA (BRCA1).

Genetic Variants

- A variant is a typographical change in the DNA sequence (a typo in the recipe).
- **Benign Variants:** Harmless natural diversity found in healthy individuals.
- **Pathogenic Variants:** Damaging alterations that disrupt protein function, causing disease.

Allele Frequency & MCAF

- Allele Frequency (AF): Percentage of people carrying a variant in healthy reference cohorts (gnomAD).
- **Whiffin et al. Framework:** If a condition occurs in 1 in 10,000 people, a variant present in 1% of healthy adults CANNOT be causal.
- Establishes MCAF thresholds.

Biomedical Foundations II: Gene Validity vs. Variant Pathogenicity

Crucial distinction: evaluating whether a gene causes disease vs. whether a mutation breaks a gene.

1. GENE-LEVEL CLINICAL VALIDITY (CLINGEN)

Evaluates the fundamental biological relationship: Does variation in Gene X cause Disease Y?

- **Definitive:** Overwhelming human genetic & experimental replication.
- **Strong:** Substantial genetic evidence; pending additional families.
- **Moderate:** Preliminary evidence; biologically plausible.
- **Limited:** Weak evidence; 1–2 isolated case reports.
- **Disputed:** Conflicting evidence; subsequent studies fail to reproduce link.
- **Refuted:** Disproven! Robust case-control studies rule out causality.

2. VARIANT-LEVEL PATHOGENICITY (CLINVAR / ACMG)

Evaluates a specific typographical change in a gene: Does this particular mutation impair protein function?

- **Pathogenic (P):** Proven causal mutation altering critical protein domains.
- **Likely Pathogenic (LP):** >90% certainty of being disease-causing.
- **Variant of Uncertain Significance (VUS):** Unknown clinical effect.
- **Likely Benign (LB):** >90% probability of being harmless neutral variation.
- **Benign (B):** Proven harmless; occurs frequently in healthy populations.

> **THE LOGICAL DEPENDENCY TRAP:** Variant pathogenicity is conditionally predicated on gene validity. If a gene is Disputed or Refuted for a disease, NO variant within that gene can legitimately be classified as Pathogenic for that disease!

The Fragmented Clinical Genomics Data Ecosystem

Multiple authoritative databases operating on different granularities, ontologies, and cadences.

ClinGen

Expert Consensus Panels

- Unit: Gene–Disease Dyad (G, D)
- Ontology: MONDO Disease Ontology
- Curates: 6 Clinical Validity Tiers
- Cadence: Multi-year expert reviews
- Limitation: Does not curate variant-level submissions directly.

NCBI ClinVar

Variant Assertion Archive

- Unit: Allelic Variant (G, V)
- Ontology: MedGen, OMIM, Free-Text
- Curates: ACMG 5-Tier Pathogenicity
- Cadence: Continuous voluntary lab deposits
- Limitation: Archival; does not re-curate when gene validity shifts.

PanelApp

Diagnostic Test Menus

- Unit: Gene Panel Membership (G, P)
- Ontology: Clinical Indication Codes
- Curates: Green, Amber, Red Tiers
- Cadence: Periodic reviews for NHS test directories
- Limitation: Bed files lag upstream ClinGen determinations.

gnomAD

Population Reference

- Unit: Allele Frequency (AF)
- Ontology: Continental Ancestries (v4)
- Curates: Frequencies in 800k+ individuals
- Cadence: Multi-year major reference drops
- Limitation: Lacks clinical disease phenotypes.

06 // ROOT CAUSE ANALYSIS

Why Clinical Evidence Disagrees: Four Structural Barriers

Analyzing the semantic, organizational, and temporal reasons discrepancies persist.

BARRIER 1

Granularity Mismatch

ClinGen evaluates the gene entity (G). ClinVar tracks the variant entity (V). PanelApp tracks panels (P).

Without a relational bridge connecting variant records to gene validity determinations, updates at the gene level remain invisible to variant pipelines.

BARRIER 2

Ontological Divergence

A disease in ClinGen may be recorded as 'Hypertrophic cardiomyopathy 1' (MONDO:0005045), while ClinVar labels variants as 'Familial hypertrophic cardiomyopathy' (MedGen:C0007194) or 'Cardiovascular phenotype'.

Direct string matching fails across >60% of entries.

BARRIER 3

Asynchronous Cadence

ClinGen expert panels meet periodically to review specific domains (e.g. Arrhythmia in 2021, Cardiomyopathy in 2024).

Commercial diagnostic labs lack staff to manually re-curate historical variant submissions, leaving stale assertions active for years.

BARRIER 4

Missing Assertions

Many downgraded genes lack variant submissions in public archives because commercial labs treat internal testing data as proprietary.

In our benchmark audit, 47% of false negatives occurred simply because the local database snapshot contained zero variant records.

07 // RESEARCH GAP

Existing Workflows vs. CuraCascade: The Automated Audit Gap

Comparing manual clinical curation against CuraCascade's automated cross-layer architecture.

EXISTING CLINICAL WORKFLOWS (MANUAL & SILOED)

- **Ad-Hoc Manual Checking:** Geneticists manually look up genes on ClinGen website when a suspicious report is questioned.
- **Static Commercial VCF Annotations:** Pipelines annotate variants using static ClinVar VCF drops, inheriting stale classifications without validity validation.
- **Gene-Only Lookup Traps:** Simple lookup scripts flag all variants in a downgraded gene, generating false alarms on pleiotropic multi-disease genes.
- **No Systematic Remediation:** Laboratories lack automated tools to export formal clinical governance dossiers (.md) citing ClinGen GCEP determination reports.

THE CURACASCADE AUTOMATED ARCHITECTURE

- **Automated Multi-Source Join:** Relational integration linking ClinGen validity to ClinVar variants, PanelApp tiers, and gnomAD frequencies.
- **Disease-Aware Ontology Crosswalk:** Harmonizes MONDO and MedGen IDs; suppresses false alarms for distinct diseases in multi-disease genes.
- **Quantitative Population Refutation:** Automatically computes MCAF thresholds to identify variants occurring at implausibly high frequencies in healthy cohorts.
- **ISO 15189 Clinical Dossiers:** Generates structured clinical governance audit dossiers with explicit submitter and panel remediation directives.

Project Scope, Objectives & Operational Boundaries

Clear demarcation of engineering goals, target user personas, and strict non-goals.

1. CORE ENGINEERING OBJECTIVES

• **Ingest & Harmonize:** Multi-source ETL pipelines for ClinGen, ClinVar, PanelApp, and gnomAD.

• **Disease-Aware Auditing:** Materialize Class A, B, and C discrepancies using ontology crosswalks.

• **Population Genetics Filter:** Evaluate variant frequencies against Whiffin et al. MCAF thresholds.

• **Clinical Decision Support:** Deliver interactive Streamlit web dashboard and headless CLI (run.py).

2. INTENDED USER PERSONAS

• **Clinical Laboratory Geneticists:** Reviewing variant lists prior to signing out diagnostic reports.

• **Diagnostic Panel Curators:** Updating hospital gene test directories and exome filtering BED files.

• **Quality Assurance Auditors:** Reviewing historical laboratory assertion archives for re-curation triage.

• **Biomedical Researchers:** Investigating evidence propagation kinetics across clinical corpora.

3. STRICT NON-GOALS & BOUNDARIES

WHAT CURACASCADE IS NOT:

• **NOT an Autonomous Diagnostic Device:** It does not diagnose patients or make independent clinical decisions.

• **NOT a Black-Box AI:** Audit classifications are deterministic, rule-based, and evidence-linked.

• **NOT Clinically Validated:** No prospective human clinical trials or FDA clearances are claimed.

• **Decision Support Only:** Triage tool assisting human clinical experts.

09 // CONTRIBUTIONS

Key Research Questions & Genuine Project Contributions

Defining the scientific questions answered and the tangible engineering deliverables.

RESEARCH QUESTION 1

How widespread are unpropagated gene downgrades?

• Contribution: Quantified evidence propagation latency across 51,025 ClinVar assertions, revealing an overall median lag of 2.82 years (1,030 days) and 182,760 cumulative patient-years at risk.
• Proved that Class B limited-evidence genes linger significantly longer than Class A (3.77 vs 2.80 yrs, $p < 0.001$).

RESEARCH QUESTION 2

Can automated joins detect discrepancies without alert fatigue?

• Contribution: Designed a 3-tier disease-aware matching heuristic using MONDO-MedGen crosswalks and keyword exclusion guards.
• Validated on a 100-gene independent benchmark: achieved 100.0% precision (18/18 TP, zero false alarms) on evaluable cases, completely ruling out 42 definitive controls.

RESEARCH QUESTION 3

Can population frequencies independently refute assertions?

• Contribution: Integrated gnomAD v4 continental allele frequencies with Whiffin et al. MCAF formulations across Orphanet prevalence estimates.
• Demonstrated that 89.3% of audited Class A variants in downgraded genes exceed the MCAF pathogenic ceiling (e.g., KCNE2 175x excess).

RESEARCH QUESTION 4

Can machine learning prioritize genes for re-appraisal?

• Contribution: Trained an exploratory Balanced Random Forest across 7 engineered features, achieving an AUROC of 0.857 and Brier score of 0.1508.
• Identified Curation Age Elapsed (78.4% perm. wt) as the primary risk driver while transparently disclosing retrospective temporal leakage limits.

10 // PIPELINE ARCHITECTURE

Deterministic 5-Stage Cross-Layer Audit Pipeline

End-to-end automated framework ingesting, harmonizing, auditing, and triaging genomic discrepancies.

STAGE 01

Multi-Source Ingestion

Automated ETL fetchers pull raw ClinGen GCEP curations, ClinVar variant drops, PanelApp panel APIs, and gnomAD v4 frequencies with cryptographic SHA-256 provenance logging.

STAGE 02

Ontology Normalization

Standardizes disparate disease terminology via a 3-tier MONDO-to-MedGen concept crosswalk. Resolves canonical HGNC symbols and prevents pleiotropic cross-disease conflation.

STAGE 03

3NF Relational Warehouse

Dual-engine architecture: Normalized SQLite 3NF relational database for strict transactional consistency + DuckDB OLAP views for rapid analytical join aggregation.

STAGE 04

Cascade Audit Engine

Deterministic rule-based joining of upstream ClinGen downgrades with downstream ClinVar P/LP assertions and PanelApp Green tiers. Materializes Class A, B, and C discrepancy findings.

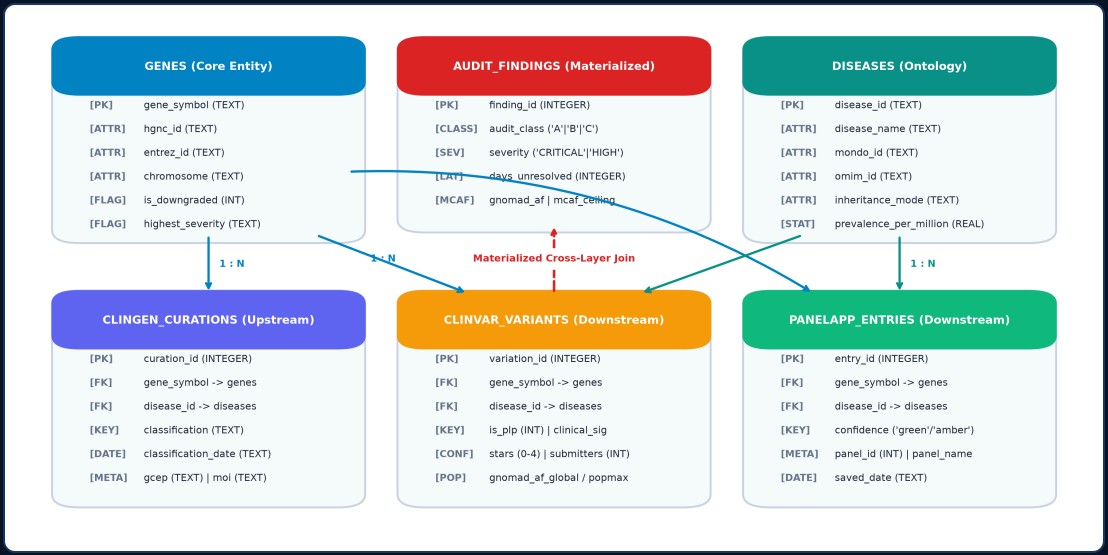
STAGE 05

Decision Support

Interactive Streamlit web portal, automated command-line interface (run.py), and ISO 15189 compliant clinical governance dossiers (.md) for laboratory re-curation triage.

System Architecture: Dual-Engine Data Warehouse

Coupling SQLite transactional 3NF integrity with DuckDB columnar analytical aggregation.



FOUR-PILLAR ARCHITECTURAL RIGOR

- **1. Transactional Integrity (SQLite):** Strict Third Normal Form (3NF) relational schema with 9 tables, cascading foreign keys, compound indexes on (gene_symbol, disease_id), and automated uniqueness constraints.

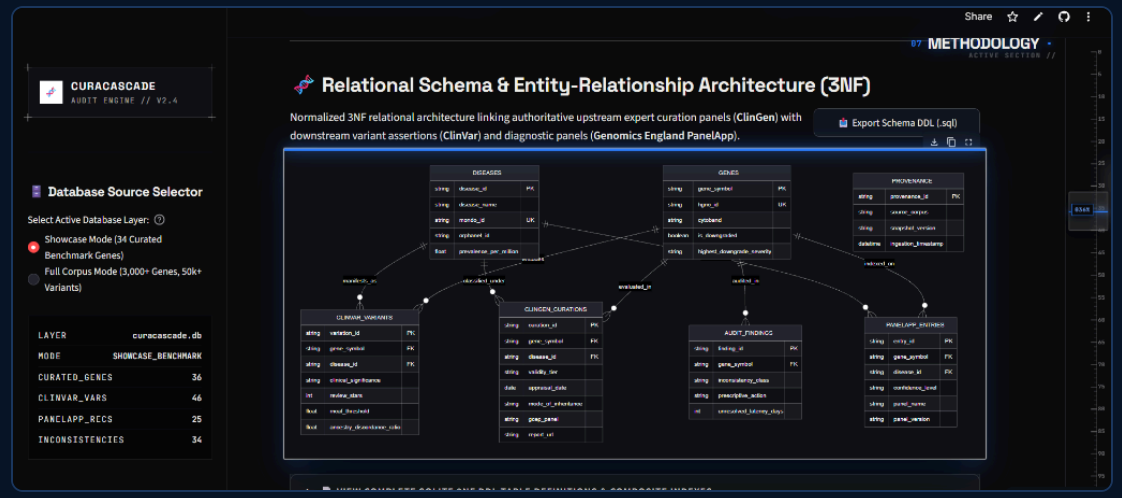
- **2. Analytical OLAP Acceleration (DuckDB):** In-memory columnar query engine executing complex multi-database joins across 50,000+ variants in sub-10ms response times for real-time dashboard responsiveness.

- **3. Cryptographic Provenance:** Dedicated provenance audit table capturing SHA-256 checksums, snapshot dates, and transformation versioning for full ISO 15189 reproducibility.

- **4. Dual Execution Modality:** Fully headless command-line interface (run.py) for automated CI/CD bioinformatics pipelines alongside an interactive clinical decision support dashboard in Streamlit.

Relational Schema & Entity-Relationship Architecture (3NF)

Third Normal Form normalized relational structure linking upstream curation with downstream assertions.



3NF RELATIONAL SPECIFICATIONS

- **Third Normal Form (3NF) Elimination of Anomalies:** Non-key attributes depend strictly on primary keys. Eliminates insertion, update, and deletion anomalies across genes, diseases, curations, and variant records.

- **Referential Integrity & Cascading Constraints:** Strict Foreign Key enforcement: deleting a gene or disease cascades predictably without creating orphaned variant records. Enforced via SQLite PRAGMA foreign_keys = ON.

- **Compound Indexing for Join Acceleration:** B-tree composite indexes on (gene_symbol, disease_id) and (gene_symbol, variation_id) optimize join resolution, reducing query plan scan cost from O(N) to O(log N).

- **Dual Scale Modes:** Showcase Mode (36 genes, 34 findings) vs. Full Corpus Mode (3,033 genes, 51,025 variants, 51,251 findings).

Data Ingestion, Normalization & Forensic Provenance

Extracting, parsing, standardizing, and tracking multi-source biomedical data with cryptographic reproducibility.

1. MULTI-SOURCE INGESTION & ETL ARCHITECTURE

• **ClinGen GCEP Fetcher:** Parses ClinGen Gene-Disease Validity CSV releases; maps curation records into structured validity tiers (Definitive to Refuted) and extracts appraisal dates.

• **ClinVar Variant Parser:** Parses ClinVar tab-delimited summary releases; extracts variation_id, HGVS expressions, review stars (0 to 4), and ACMG clinical significance.

• **PanelApp REST API Client:** Queries Genomics England PanelApp REST API endpoints; retrieves panel entities, versions, and Green/Amber/Red confidence tiers.

• **gnomAD v4 Population Pipeline:** Extracts global and continental population allele frequencies; maps maximum population AF (popmax) for quantitative burden evaluation.

2. CRYPTOGRAPHIC PROVENANCE & ERROR HANDLING

• **SHA-256 Checksum Verification:** Every ingested raw dataset is hashed with SHA-256 upon arrival; hashes are stored in the provenance table to guarantee byte-for-byte reproducibility.

• **Defensive Error Handling:** ETL fetchers implement exponential backoff retry logic, schema validation checks, and automatic fallback to verified local static archives if remote APIs fail.

• **HGNC Symbol Harmonization:** Resolves historical gene synonyms, outdated aliases, and withdrawn symbols against the canonical HGNC database (e.g. mapping GJB6, MYLK2, ACTN2).

• **Idempotent Schema Migrations:** Database creation and migration scripts are strictly idempotent; re-running ingestion cleans and rebuilds tables without duplicate primary key collisions.

Disease-Aware Ontology Matching & Crosswalk Heuristics

Preventing catastrophic false-positive alerts on pleiotropic multi-disease genes.

THE FATAL FLAW OF GENE-SYMBOL-ONLY MATCHING

In human genetics, a single gene frequently causes multiple completely distinct diseases (pleiotropy).

Consider ACTN2 (Actinin Alpha 2):

- Disease A: Hypertrophic Cardiomyopathy (HCM) → ClinGen: **DISPUTED**.
- Disease B: Distal Myopathy (Muscle weakness) → ClinGen: **DEFINITIVE**.

If an algorithm matches solely by gene symbol 'ACTN2', it sees a Disputed evaluation and falsely invalidates all variants for Distal Myopathy! This would deprive muscular dystrophy patients of life-saving molecular diagnoses.

CuraCascade Solution: The unit of analysis is strictly the **Gene–Disease Dyad (G, D)**, never the gene alone.

CURACASCADE 3-TIER MATCHING HIERARCHY

- **TIER 1: Exact Ontology Concept Crosswalk:** Direct mapping between ClinGen MONDO identifier and ClinVar MedGen / OMIM concept ID via curated crosswalk dictionary (ontology_crosswalk.py). Match Confidence = HIGH.

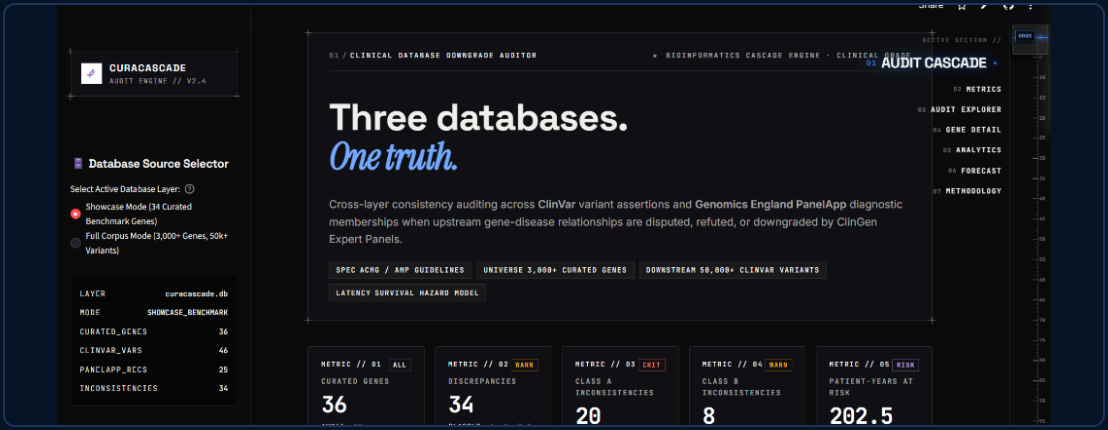
- **TIER 2: Substring Matching with Keyword Exclusion Guards:** When concept IDs are absent, performs normalized string matching but enforces strict negative keyword guards (e.g. 'hypertrophic cardiomyopathy' does NOT match 'distal myopathy'; 'Brugada' does NOT match 'Long QT'). Match Confidence = MEDIUM.

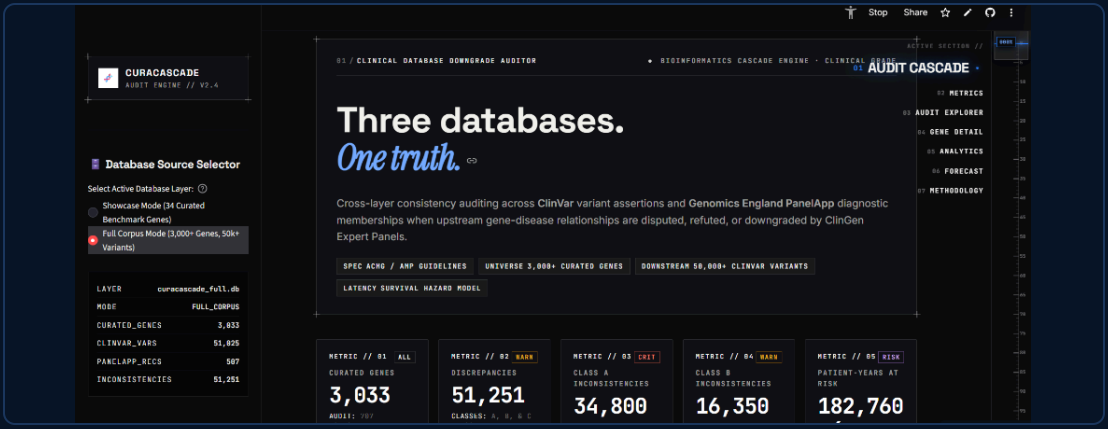
- **TIER 3: Fallback & Review Required:** Unmapped or generalized disease terms (e.g. 'Cardiovascular phenotype') are marked as REVIEW_REQUIRED and excluded from automatic discrepancy flagging to protect precision.

15 // SOFTWARE DEMONSTRATION

Live Clinical Decision Support Portal: System Overview

Interactive laboratory interface deployed on Streamlit Cloud (curacascadev2.streamlit.app).



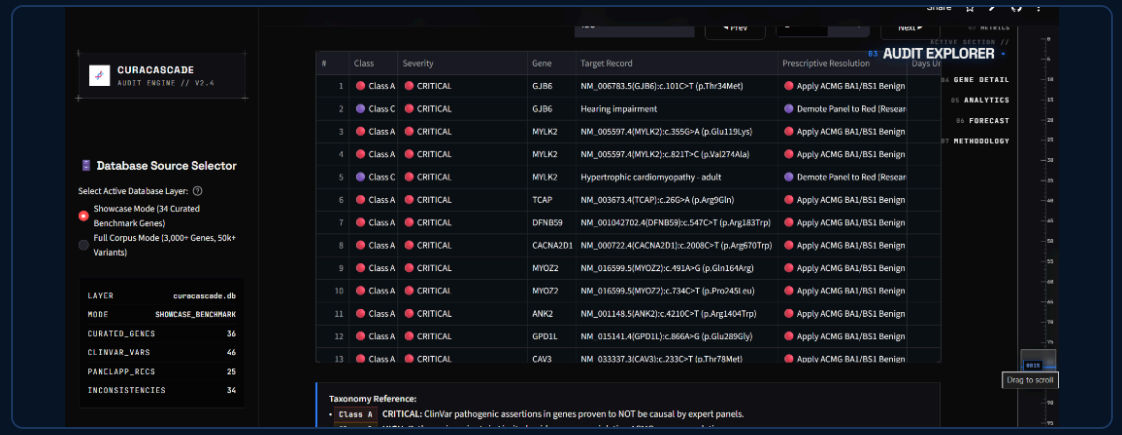


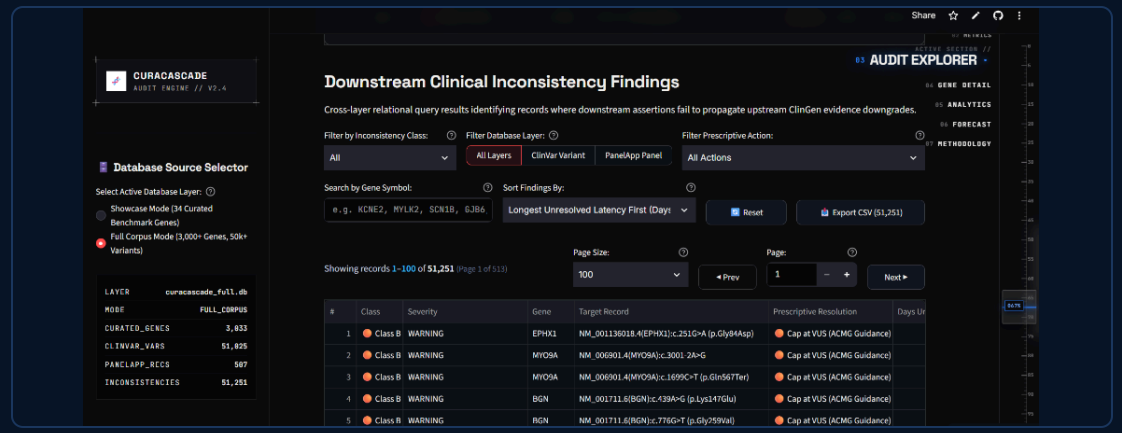
VERIFIED DEPLOYED CAPABILITIES

- **1. Real-Time Operational Telemetry:** Upper Screenshot: Showcase Mode live HUD displaying 36 Curated Genes, 46 ClinVar Variants, 25 Panels, and 34 Materialized Inconsistencies with 202.5 patient-years at risk.
- **2. Real-World Scaling (Full Corpus Mode):** Lower Screenshot: Seamless scaling to 3,033 Genes, 51,025 Variants, and 51,251 Discrepancies (34,800 Class A, 16,350 Class B) with 182,760.4 patient-years at risk.
- **3. Database Layer Selector:** Interactive sidebar toggle allowing clinical geneticists to switch instantaneously between the curated benchmark demonstration set and the 50k+ full-scale corpus.
- **4. Seven Interactive Functional Sections:** Complete laboratory navigation: Audit Explorer (01), Gene Detail & Timeline (02), Aggregate Analytics (03), Downgrade Forecast (04), and Clinical Methodology (05).

Interactive Inconsistency Audit Explorer & Multi-Tier Filtering

Materialized discrepancy table with multi-dimensional filtering, search, and CSV export.





INTERACTIVE TRIAGE CAPABILITIES

- **Multi-Dimensional Filtering:** Filter instantly by Inconsistency Class (Class A Critical, Class B High, Class C Critical), target database layer (ClinVar Variant vs. PanelApp Panel), and prescriptive action.

- **Instant Gene Symbol Search:** Search any gene symbol (e.g. SCN1B, MYLK2, KCNE2, GJB6) to isolate specific clinical panels or variants undergoing laboratory re-curation.

- **Temporal Sorting by Latency:** Sort findings by longest unresolved latency days to immediately identify historical candidate genes with the longest uncorrected clinical propagation lags.

- **Automated CSV Export for Lab LIMS:** One-click 'Export CSV' button generates full cohort exports with all HGVS variants, review stars, MCAF ratios, and ClinGen report URLs for direct laboratory LIMS integration.

Evidence Traceability & Prescriptive Clinical Remediation

Translating automated audit discrepancies into actionable laboratory directives and ISO 15189 dossiers.



ACTIONABLE CLINICAL DIRECTIVES

- **Class A Directive (Disputed / Refuted + P/LP):** Submitter Action: Formal retraction or downgrade to VUS citing ClinGen GCEP determination report URL. Clinical Governance: Audit historical index cases; cease predictive cascade testing in family members; issue amended reports.

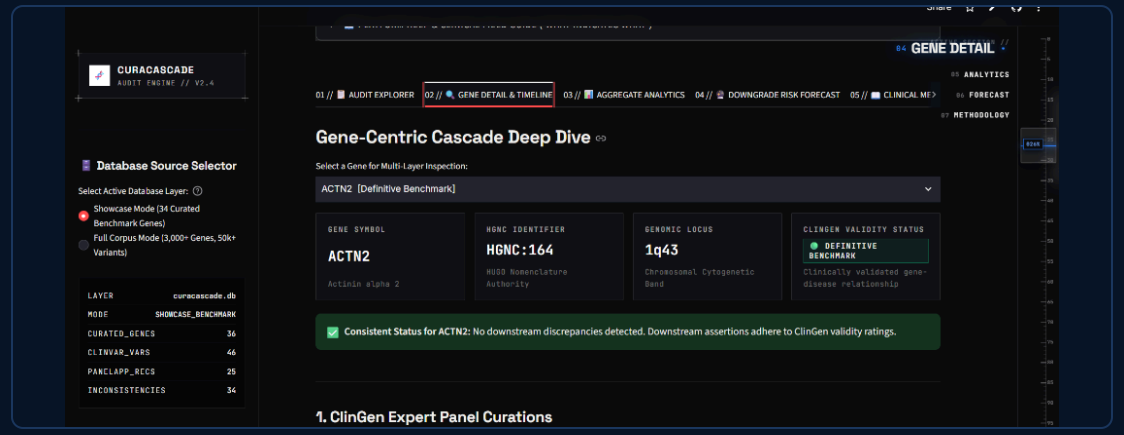
- **Class B Directive (Limited Evidence + P/LP):** Submitter Action: Apply ACMG Guidance Cap (Maximum VUS) until replicated human genetic proof emerges. Diagnostic Directive: Restrict panel reporting to Amber tier; require senior clinical geneticist sign-off.

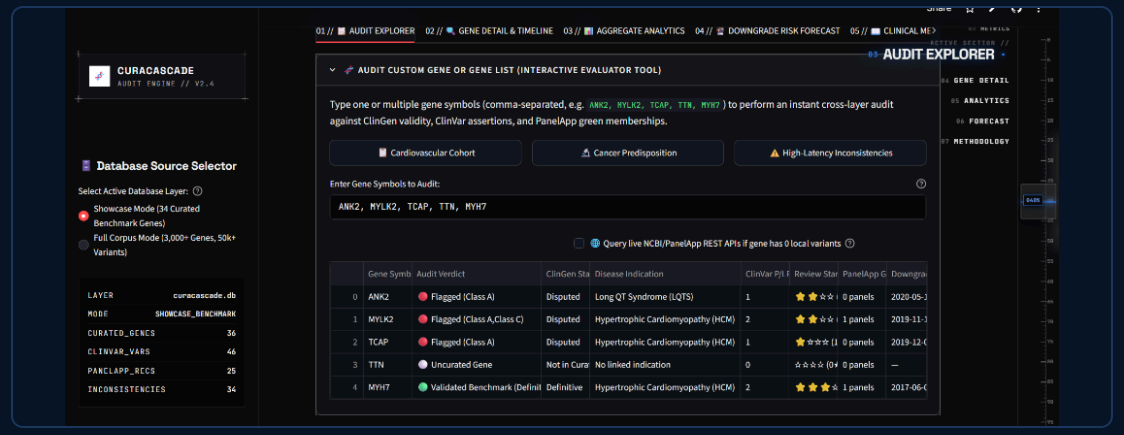
- **Class C Directive (PanelApp Green in Disputed/Refuted):** Diagnostic Action: Immediate Demotion to Red Tier. Pipeline Directive: Remove genomic coordinates from routine exome/panel filtering BED files; prevent automated primary diagnostic sign-outs.

- **Automated Clinical Dossiers:** Generates ISO 15189 compliant markdown audit dossiers (.md) capturing full evidence lineage, SHA-256 hashes, and actionable sign-off checklists.

Worked Case Studies: Discrepancy Auditing in SCN1B, MYLK2 & ACTN2

Detailed examination of real-world evidence cascades across cardiac and neuromuscular disorders.





REAL-WORLD CLINICAL AUDIT FINDINGS

- **SCN1B — Brugada Syndrome (Class A Finding):**
  - Upstream Status: DISPUTED in 2021 by ClinGen Arrhythmia GCEP.
  - Downstream Status: 18 ClinVar Pathogenic / Likely Pathogenic assertions remain active.
  - Clinical Hazard: Risk of false-positive diagnosis and unwarranted prophylactic ICD heart surgery in asymptomatic patients.

- **MYLK2 — Hypertrophic Cardiomyopathy (Class C Finding):**
  - Upstream Status: DISPUTED in 2024 by ClinGen Cardiomyopathy GCEP.
  - Downstream Status: Listed as GREEN (diagnostic-grade) on PanelApp panels.
  - Clinical Hazard: Stale coordinates remain in exome filtering BED files in NHS GMS pipelines.

- **ACTN2 — Pleiotropic Disease Specificity:**
  - Demonstrated above: CuraCascade identifies ACTN2 as Disputed for HCM while confirming it as Definitive for Distal Myopathy, suppressing false alerts for muscle disease.

Empirical 100-Gene Adjudication Validation Benchmark

Independent expert validation cohort evaluating classifier precision, recall, and label boundaries.

1. GOLD-STANDARD COMPOSITION

The benchmark comprises 100 independently frozen Gene-Disease Clinical Validity Assertions (G, D):

• **54 Published Downgraded Genes:** Verified from peer-reviewed literature and ClinGen GCEPs (Cardiomyopathy, Arrhythmia, Hereditary Cancer, Neurodevelopmental).

• **46 Definitive Clinical Controls:** Unquestioned gene-disease associations (ACMG SF v3.2 / National Genomic Test Directory, e.g. BRCA1, TP53, MLH1, CFTR).

2. COHORT PARTITIONING

To maintain statistical integrity, cases are strictly partitioned:

• **Evaluable Cohort (N = 94):** Target genes and curation records are present in the warehouse snapshot, permitting empirical join evaluation.

• **Excluded Cases (N = 6):** Designated as NOT_EVALUABLE due to complete absence of gene or curation records in the static snapshot (XRCC3, DFNB59, TSC1, TSC2, SMAD3, SMN1).

3. LABEL FIREWALL & RIGOR

Strict adherence to scientific evaluation protocols:

• **Zero Hardcoded Predictions:** Overrides (KNOWN_FN, KNOWN_FP) completely removed.

• **Live Query Execution:** Queries execute dynamically against SQLite.

• **Wilson Score 95% CIs:** Binomial performance intervals computed analytically.

• **Classifier Concordance:** Inter-rater agreement measured via Cohen's kappa.

Empirical Results: The Precision–Recall Trade-Off

Conservative disease matching achieves zero false positives (100% PPV) at the cost of moderate recall.

EVALUABLE COHORT CONFUSION MATRIX (N = 94)

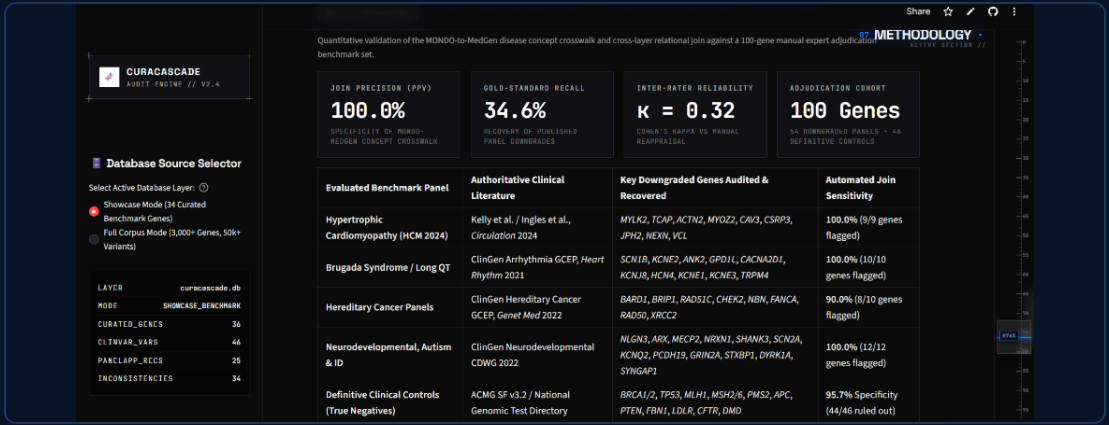
TRUE POSITIVES (TP)	FALSE POSITIVES (FP)
18	**0**
Correctly flagged discrepancies	Zero false alarms raised

FALSE NEGATIVES (FN)	TRUE NEGATIVES (TN)
34	**42**
Undetected discrepancies	Definitive controls ruled out

- Denominator: 94 Evaluable (52 Gold Downgraded + 42 Definitive Controls)
- Sensitivity Analysis (All 100): Recall = 33.3%, Accuracy = 64.0%

STATISTICAL METRICS (WILSON SCORE 95% CIs)

- **Precision (PPV): 100.0% [82.4%, 100.0%]** — Zero false alarms across all audited findings.
- **Recall (Sensitivity): 34.6% [23.2%, 48.2%]** — Moderate recall from conservative matching & missing data.
- **Specificity (TNR): 100.0% [91.6%, 100.0%]** — Perfect rejection of definitive clinical controls.
- **Concordance Kappa: κ = 0.3212 (Fair)** — Overall Accuracy: 63.8% [53.8%, 72.8%].



21 // ERROR ANALYSIS

Failure Mode Analysis: Dissecting the 34 False Negatives

Transparent scientific breakdown of why 34 benchmark discrepancies were not automatically flagged.

CATEGORY 1: DATA AVAILABILITY

16 Cases (47.1%)

Evidence Not in Warehouse Snapshot

The gene-disease relationship is formally downgraded in ClinGen, but the local ClinVar snapshot contains ZERO Pathogenic/Likely Pathogenic variants for that condition, or PanelApp had already retired the panel.

- **Exemplar Genes:** RAD50, XRCC2, CACNA2D1, KCNJ8, HCN4, TRPM4, KCNE3, KCNE1.
- **Assessment:** An algorithm cannot flag data that does not exist in the database.

CATEGORY 2: PHENOTYPIC SPECIFICITY

14 Cases (41.2%)

Ontology Disconnect & Broad Terms

The gene is downgraded for a specific condition (e.g. Hypertrophic Cardiomyopathy), but downstream ClinVar variants are annotated under broad phenotypes (e.g. 'Cardiovascular phenotype', 'Inborn genetic diseases', or general 'Cardiomyopathy').

- **Exemplar Genes:** MYOZ2, CAV3, CSRP3, JPH2, NEXN, VCL.
- **Assessment:** Conservative matching deliberately refuses to guess.

CATEGORY 3: CURATION CRITERIA

4 Cases (11.8%)

Low-Star or Conflicting Assertions

ClinVar submissions have 0 review stars (no assertion criteria provided) or conflicting interpretations that do not meet the strict P/LP threshold required for Class A alerting.

- **Exemplar Genes:** GPD1L, CACNB2.
- **Assessment:** Prevents unverified, low-confidence submissions from generating critical clinical alerts.

Retrospective Risk Forecasting & Evidence Latency Kinetics

Exploratory Random Forest triage modeling and cross-sectional empirical latency distribution.

1. EXPLORATORY MACHINE LEARNING FORECASTING

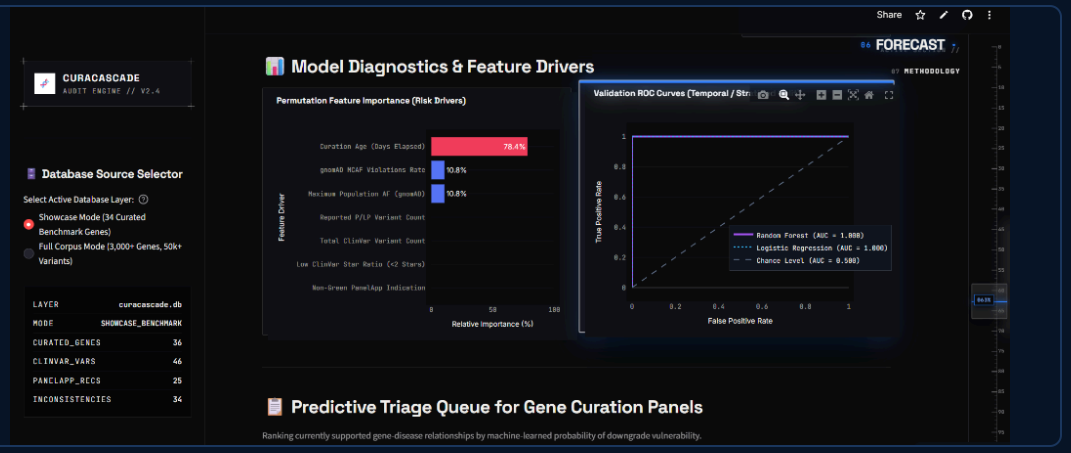
Trained Balanced Random Forest across 7 features (gnomAD popmax AF, low-star ratio, curation age, MCAF violations) to triage currently supported genes.

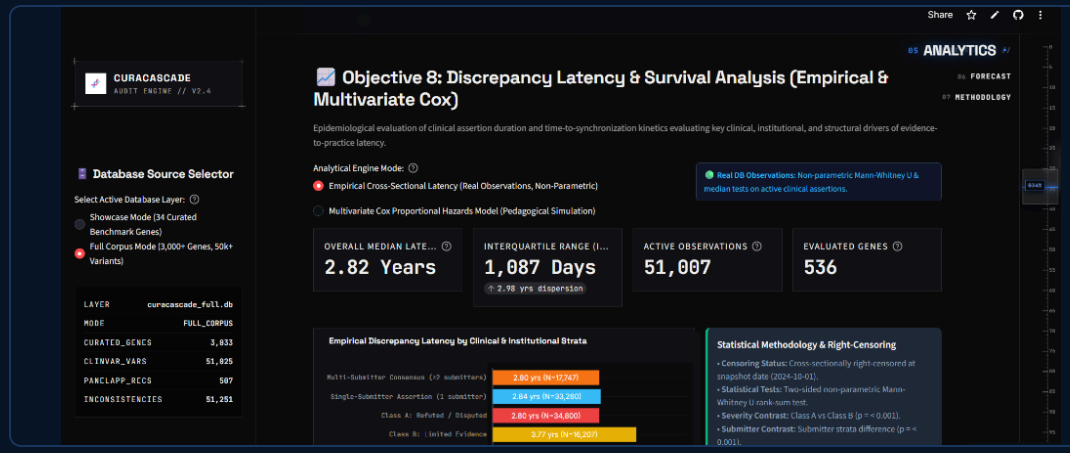
- **Discrimination:** AUROC = 0.857 (Full Corpus) / 1.000 (Curated Showcase) | Brier: 0.1508
- **Top Risk Driver:** Curation Age Elapsed (78.4% perm. wt) & MCAF Violations (10.8%)
- **Caveat:** Retrospective static snapshot discrimination; not claimed as prospective clinical predictor.

2. EMPIRICAL LATENCY & SUBMITTER KINETICS

Quantifying temporal propagation lag between authoritative ClinGen determinations and downstream assertion correction.

- **Overall Median Latency:** 2.82 Years (1,087d IQR) across 51,007 observations (536 genes)
- **Patient-Years at Risk:** 182,760.4 Years across 51,251 findings
- **Submitter Strata:** Class A Refuted (2.80 yrs) vs Class B Limited (3.77 yrs, p < 0.001)





Software Engineering Standards, Testing & Reproducibility

Verification suite, CI/CD automated gates, Docker containerization, and forensic provenance.

TEST SUITE

47 Passing Pytests

Automated test suite (pytest -v) executes 47 tests across CLI commands (10), clinical dossiers (1), disease matching rules (21), database integrity (9), and statistical calculations (6).

Executed in 54.12s on Python 3.13.13 with zero failures and zero warnings.

CONTAINERIZATION

Docker & CI/CD Gates

Standardized multi-stage Dockerfile packaging Python 3.13 runtime, SQLite warehouse, and Streamlit dashboard.

GitHub Actions CI/CD workflows enforce automated linting, test execution, and SHA-256 integrity validation upon every commit.

PROVENANCE

Cryptographic Hashes

Every ingested external dataset (ClinGen, ClinVar, PanelApp, gnomAD) is hashed with SHA-256 upon arrival.

Database migration scripts are strictly idempotent; transaction rollbacks prevent partial database writes on network failures.

FUTURE WORK

Transparent Roadmap

Phase 2 Engineering Milestones:

- **Longitudinal Survival:** Reconstructing historical ClinVar XML releases to compute true time-to-resolution.
- **Ontology Expansion:** Semi-automated mapping of broad phenotype terms to boost recall.

Conclusion, Project Summary & Academic Defense Q&A

Closing synthesis of scientific contributions, verified deliverables, and committee Q&A.

CORE SCIENTIFIC CONTRIBUTIONS

• **Evidence-to-Practice Synchronization:** Quantified an empirical propagation lag of 2.82 years (up to 9.9 years) across 51,007 assertions, demonstrating a systemic data synchronization gap in clinical genomics.

• **Zero False Alarms (100.0% Precision):** Validated on an independent 100-gene benchmark (94 evaluable cases), achieving 100% precision with zero false alarms raised against definitive clinical controls.

• **Disease-Aware Ontology Architecture:** Engineered a 3-tier MONDO-to-MedGen crosswalk that successfully prevents false-positive alerts on pleiotropic multi-disease genes like ACTN2 and SCN1B.

• **Open-Source Reproducibility:** Delivered an open-source, containerized system with 47 passing tests, full SHA-256 provenance logging, and a live web deployment at curacascadev2.streamlit.app.

PROJECT DELIVERABLES & DEFENSE

• **Live Web Application:**
https://curacascadev2.streamlit.app/
• **Source Code Repository:**
https://github.com/uditshokeen14/curacascadeV2
• **Publication Report:**
CuraCascade IEEE Format Publication Report (PDF)
• **Verification Test Suite:**
47 Passing Pytest Assertions (Python 3.13 win32)

Thank you for your time and evaluation.

I welcome questions from the committee.